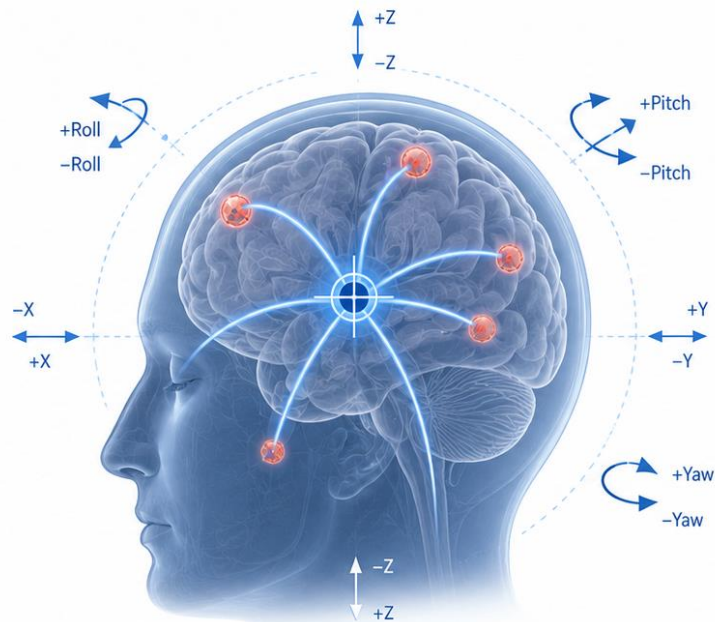
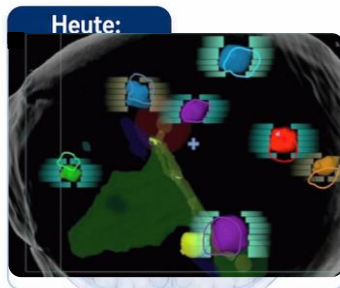
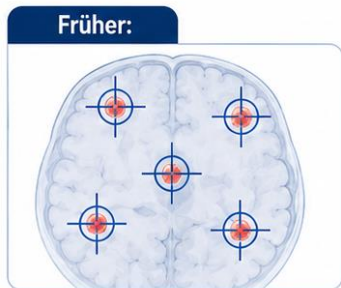




Ein Isozentrum, viele Ziele: Welche Rolle spielen intrafraktionelle Bewegungen?

Single-Isocenter-SRS bei multiplen Hirnmetastasen



6D-Positionierungsfehler
Einfluss auf Zielabdeckung
und Genauigkeit



Einflussfaktoren
Zeit, Fixierung,
Überwachung



Ringversuch
Stärkster Einfluss:
schlechte ICC;
danach fehlende
intrafraktionelle
Bildgebung



Qualitätssicherung
Normale WLT und
Off-Isocenter-WLT

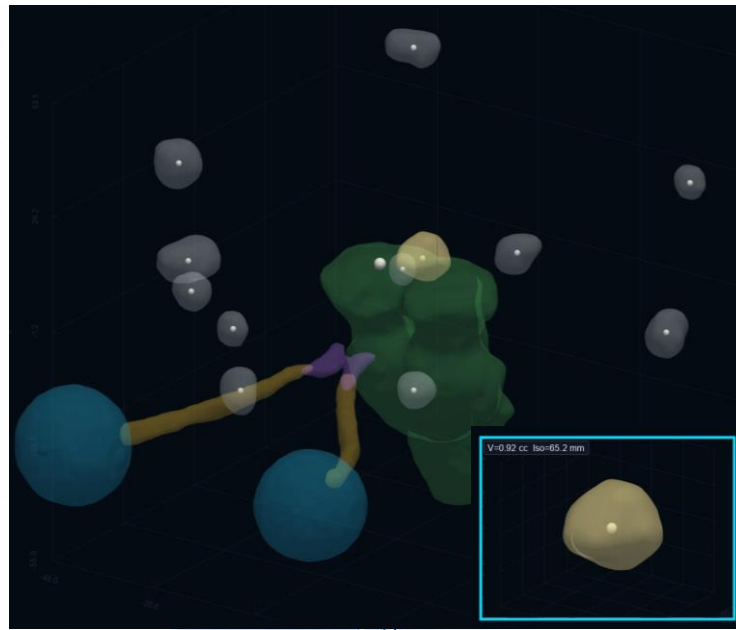
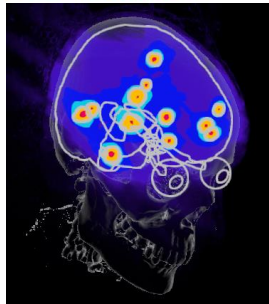
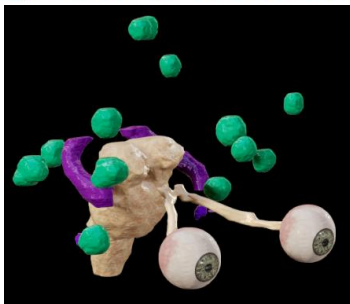


Klinischer Workflow
Nachkorrigieren und
patientennaher Ablauf



Ein Isozentrum, viele Ziele: Welche Rolle spielen intrafraktionelle Bewegungen?

Single-Isocenter-SRS bei multiplen Hirnmetastasen



6D-Positionierungsfehler

Einfluss auf Zielabdeckung
und Genauigkeit



Einflussfaktoren

Zeit, Fixierung,
Überwachung



Ringversuch

Stärkster Einfluss:
schlechte ICC;
danach fehlende
intrafraktionelle
Bildgebung



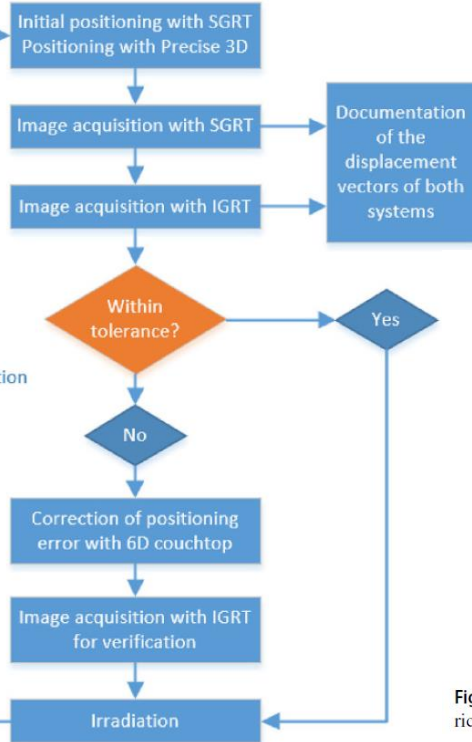
Qualitätssicherung

Normale WLT und
Off-Isocenter-WLT



Klinischer Workflow

Nachkorrigieren und
patientennahe Ablauf



Fragestellung:

6D-Korrektur?

Nach jedem CouchKick korrigieren?

Behandlungsdauer minimieren?

Margins erhöhen?

SGRT only?

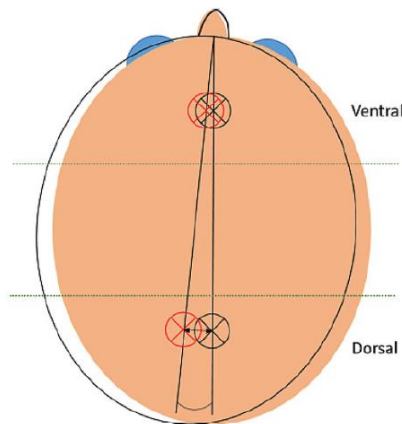


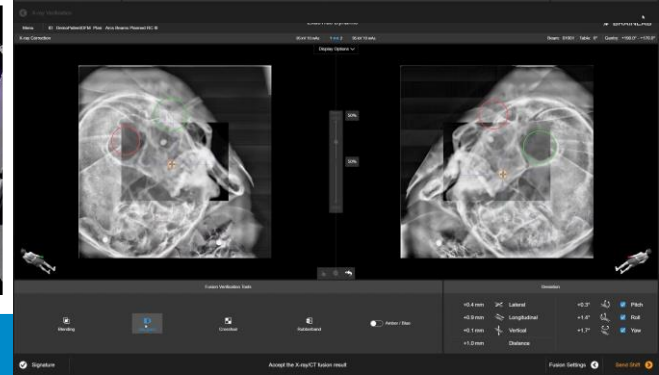
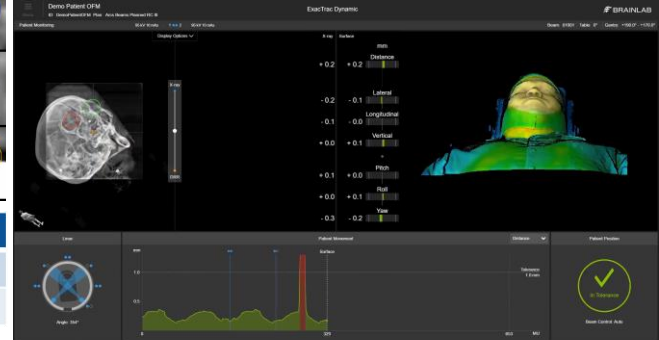
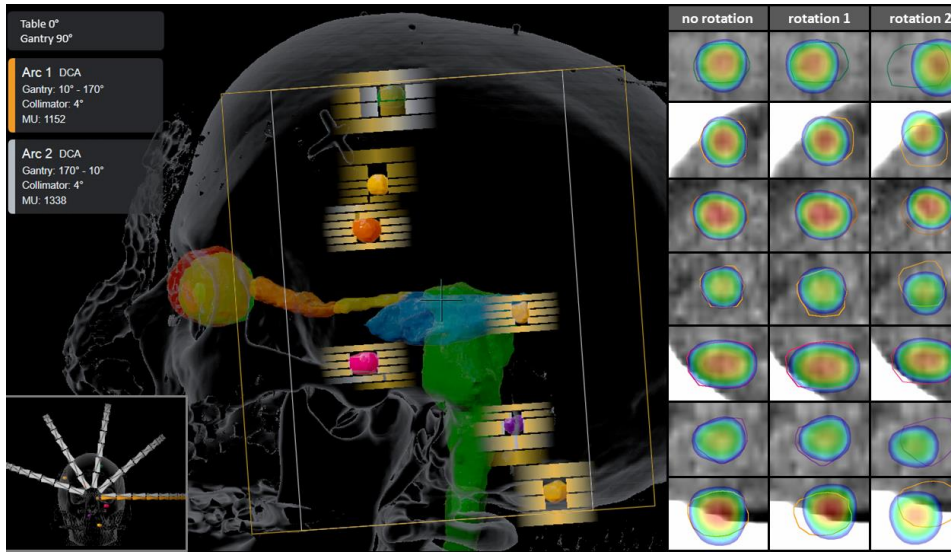
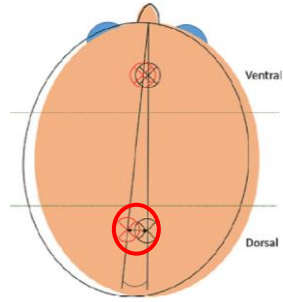
Fig.9 Illustration of rotational measurement errors and their geometric consequences depending on the face-to-isocenter distance



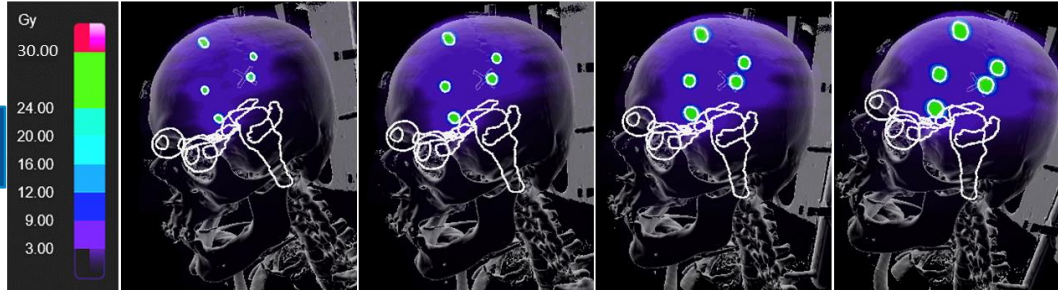
Fig.2 Data acquisition workflow using the surface- and image-guided radiation therapy systems (SGRT and IGRT)

Schöpe, M *et al.* Comparison of patient setup accuracy for optical surface-guided and X-ray-guided imaging with respect to the impact on intracranial stereotactic radiotherapy. *Strahlenther Onkol* (2024).
<https://doi.org/10.1007/s00066-023-02170-x>

Fragestellung: Margins erhöhen? In Maßen



Margin	0 mm		1 mm		2 mm		3 mm	
Gesamt-Volumen [cc]	0.264	+ 0%	0.844	+ 219.7%	1.916	+ 625.8%	3.604	+ 1265.2%
V12 [cc] Normal Brain	1.994	+ 0%	3.757	+ 88.4%	6.985	+ 250.3%	10.922	+ 447.7%



24 Gy
80%-IDL

Fragestellung:

pre-correction (XC) post-correction (XV)

6D-Korrektur? **Ja**

Nach jedem CouchKick korrigieren? **Ja**

mean = 0.71 mm
mean = 0.33 mm

Korrekturen in 66% der Couch Kicks
Bei Toleranz 0.5%/0.5mm

mean = 0.58°
mean = 0.38°

metric
CI GI D98%

all volumes

0.5 mm
0.5°

0.7 mm
1°

1 mm
1°

0 mm
1°

1 mm
0°

setup error

Grohmann, M *et al.* The Effect of Intrafractional Motion on Targeting Accuracy in Frameless Single-Isocenter Multi-Target Stereotactic Radiosurgery. *Strahlenther Onkol* (2026).

<https://doi.org/10.1007/s00066-026-02565-6> (publishing in progress | coming very soon)



Fragestellung:

6D-Korrektur? **Ja**

Nach jedem CouchKick korrigieren? **Ja**

Korrekturen in 54%/45% der Couch Kicks
Bei Toleranz 0.5%/0.5mm

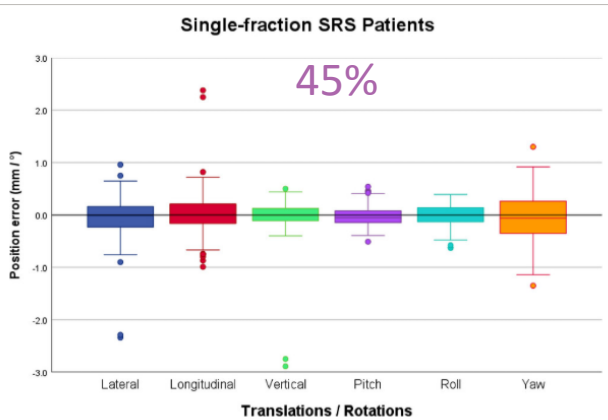


FIGURE 4

[Open in figure viewer](#)

[PowerPoint](#)

Boxplots of the distribution of the intrafraction deviations after each couch rotation during 40 SRS treatments (168 stereoscopic X-rays). The boxplots indicate the spread of the central 50% of the data, denominated as IQR. The median, the 25th (Q₁) and the 75th (Q₃) percentiles are also shown. The upper and lower whiskers represent data outside the IQR but inside the range defined by $1.5 \times \text{IQR}$. Outliers are defined as values outside the whiskers' range.

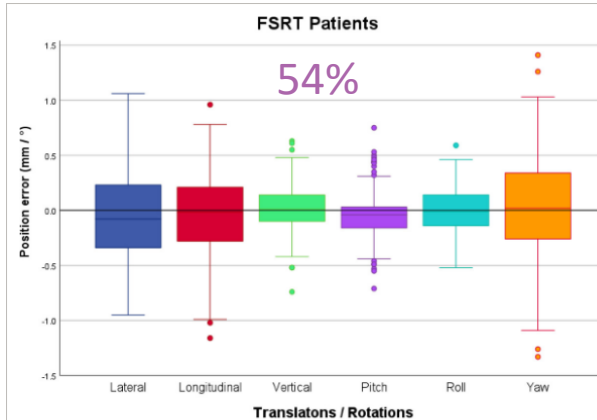


FIGURE 5

[Open in figure viewer](#)

[PowerPoint](#)

Boxplots of the distribution of the intrafraction deviations after each couch rotation during 128 FSRT fractions (278 stereoscopic X-rays). The boxplots indicate the spread of the central 50% of the data, denominated as IQR. The median, the 25th (Q₁) and the 75th (Q₃) percentiles are also shown. The upper and lower whiskers represent data outside the IQR but inside the range defined by $1.5 \times \text{IQR}$. Outliers are defined as values outside the whiskers' range.

Da Silva Mendes V et al. To shift or not to shift: identifying and correcting patient motion after couch rotations in non-coplanar intracranial radiosurgery with stereoscopic X-ray imaging. *J Appl Clin Med Phys* (2026). <https://doi.org/10.1002/acm2.70505>



Fragestellung:

Behandlungsdauer minimieren?

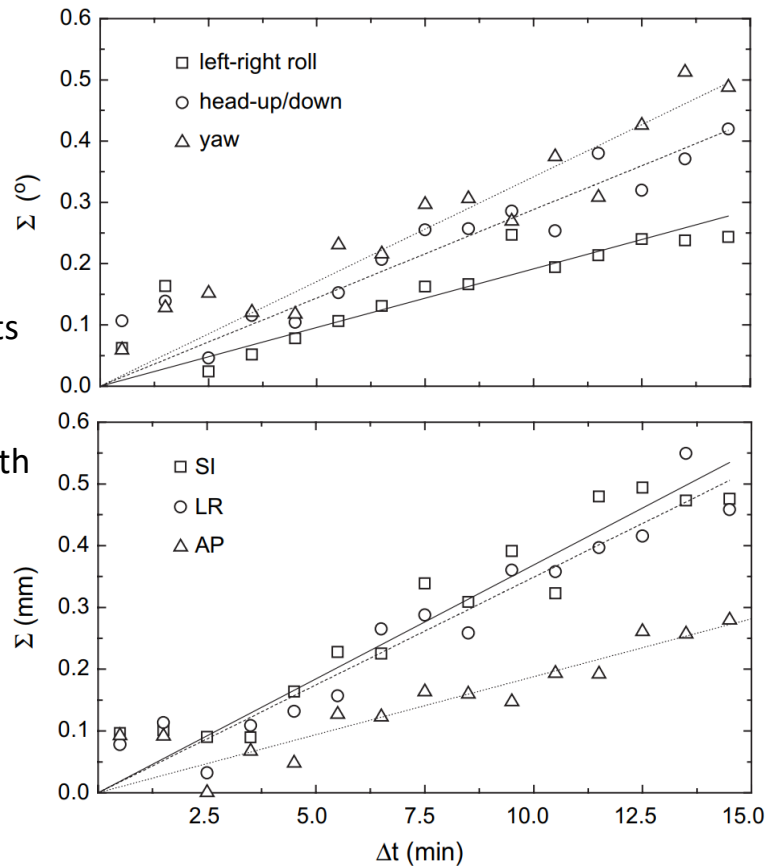
Ja

Study design:

- 32 intracranial SRS patients
- Immobilized with a thermoplastic mask
- Positioning verification with 6D-ExacTrac system

Conclusion:

- Reduction in time
= reduction in motion

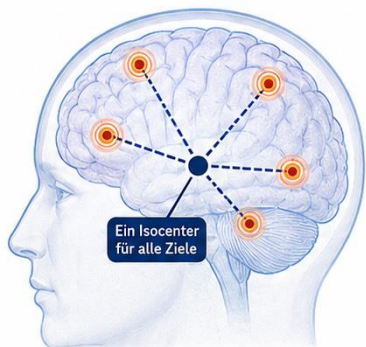


Hoogeman, Mischa S., et al. "Time dependence of intrafraction patient motion assessed by repeat stereoscopic imaging." *International Journal of Radiation Oncology* Biology* Physics* 70.2 (2008).

<https://doi.org/10.1016/j.ijrobp.2007.08.066>



Praktische Konsequenzen für die Behandlung multipler Hirnmetastasen



6D-Bewegungen

Translationen



Rotationen



Couch-Kicks (Tischrotationen)



Tischrotationen verändern die Ziel-Lage → Korrektur erforderlich



1. 6D-Korrektur?

Ja

Für präzise Zielabdeckung essenziell



2. Nach jedem Couch-Kick korrigieren?

Ja

Lagefehler nach Tischrotationen aktiv ausgleichen



3. Behandlungsdauer minimieren?

Ja

Weniger Zeit bedeutet weniger intrafraktionelle Bewegung



4. SGRT only?

Nein

Allein nicht ausreichend für höchste Präzisionsansprüche



5. Margins erhöhen?

In Maßen

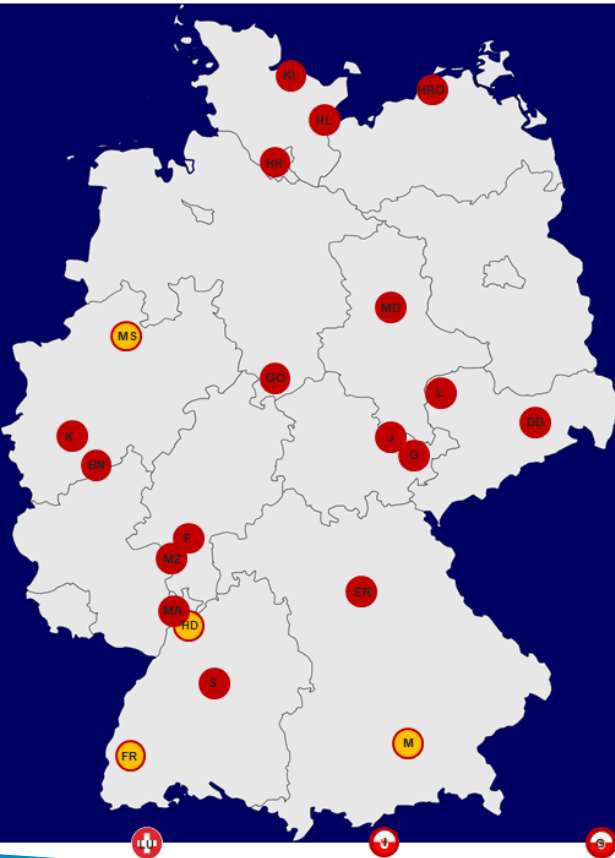
Nur begrenzt sinnvoll wegen Risiko für mehr Normalgewebe



Kernaussage: Bei SIMT-SRS sind engmaschige Bildkontrolle, Korrekturen nach Couch-Kicks und ein effizienter Workflow entscheidend.



+18 Zentren in Deutschland
+2 in Österreich
+1 in der Schweiz



Multicenter multiplatform pattern-of-practice analysis of single-isocenter multitarget stereotactic radiosurgery

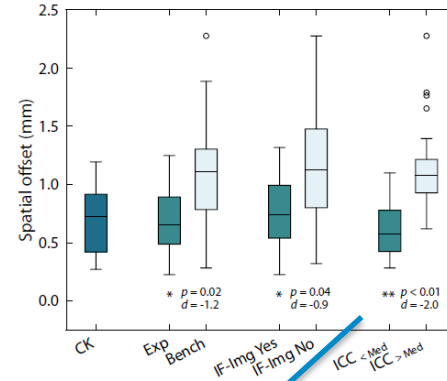
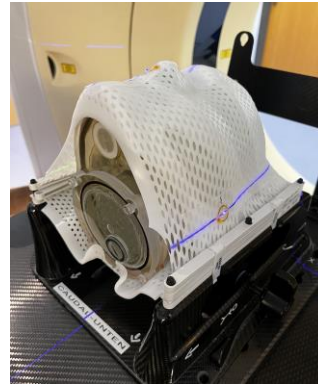


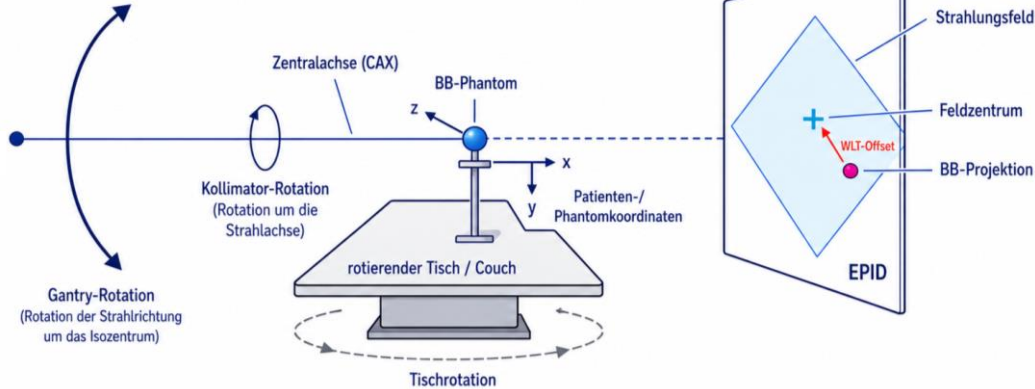
Fig. 3. Spatial offsets for all targets, analyzed for subgroups of centers: CyberKnife (CK) and C-arm centers: experienced (Exp), benchmarking (Bench), use of intrafraction imaging (IF-Img), and imaging-to-radiation isocenter consistency (ICC) level (below or above the median reported value). Boxes represent the IQR, the median (line), data range within $1.5 \times$ IQR (whiskers), and outliers beyond whiskers (points). Significance level is shown by P values, effect size magnitude by Cohen d . Asterisks indicate significance in univariate (*) or multivariate (**) tests.



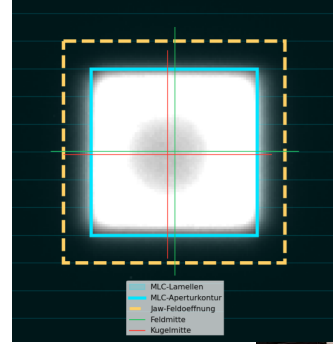
Ziel: ICC Minimierung

Lösung: ETD-Winston Lutz Test

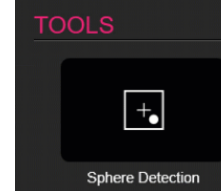
Projektion eines BB-Phantoms im Strahlungsfeld auf dem EPID



Prinzip: Der WLT vergleicht die projizierte Kugelposition mit dem Feldzentrum für verschiedene Gantry-, Kollimator- und Tischwinkel.



1. Ausrichten mit ETD



result

Deviation from isocenter [mm]
(calibrated on: 13.Mai 2026 06:52:38)

Lateral	0
Longitudinal	0
Vertical	-0.
Distance	0.
Reconstruction error	0.01

Exit



Status Acquisition Tasks Readouts

Clear Session Export Session

2. Mind. 4 MV-Aufnahmen machen

MV kV Plans

Image Mode	Type	Resolution	Acquisition Mode
Continuous	Continuous	1280x1280x16	Sync
ContinuousLowDose	Continuous	1280x1280x16	SyncLowDose
Dosimetry	Continuous	1190x1190x32	NoSyncDRI
Highres	Single	1280x1280x16	Highres
Highres-DF	Single	1280x1280x16	Highres
Highres-FF	Single	1280x1280x16	Highres
Highres_20fps	Single	1280x1280x16	Highres_20fps
Highres_20fps-DF	Single	1280x1280x16	Highres_20fps
Highres_20fps-FF	Single	1280x1280x16	Highres_20fps

MV Highres

Frames: 2

Reset Frames: 2

Dose Per Frame [MU]: 1.5

Offset Correction: ☒

Gain Correction: ☒

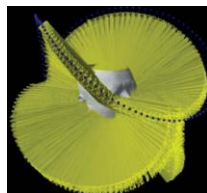
Pixel Correction: ☒



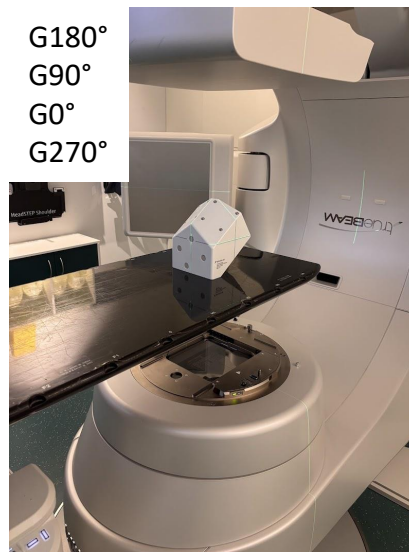
WLT TB1.dcm

RT-Plan laden für
MLC-Muster

Alternativ xml für
Komplettlauf in
DeveloperMode

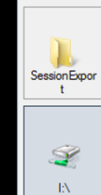


G180°
G90°
G0°
G270°



XI Service - Export Image(s)

Look in:



WLT-Plane
Backup
CBCT H191334
certs
Frederic CBCT Transfer
Gating
Gating Backup
Import
old va_transfer from varianing
QA-Plane
RTPlans
TBox-Export
TDS
WLT-Images

File name:

Save as Type

XI Image Files (*.xim)

Open

Cancel

3. Ergebnisse auswerten

ETD-WLT-UKL | pylinac v3.44.0

WINSTON-LUTZ ANALYSE

ETD-WLT-UKL | UKL Standardpfade | pylinac v3.44.0

Standard WLT

Multi-Ball Off-Iso

Verlauf

Settings

WLT MIT PYLINAC

Nutzt die aktuelle pylinac-WinstonLutz-Klasse und erzeugt optional einen PDF-Report.

Analyseordner

\\s050010021\VA_TRANSFER\WLT-Images\Tb1\2

...

.../Tb1/260616_2

keep

☒ Output im Analyseordner speichern

Allgemeiner Output-Ordner (optional)

\\medizin.uni-leipzig.de\data\Archiv\STR\STR-P1

...

.../WLT/Normal

keep

Daten

☒ TXT-Zusammenfassung schreiben

☒ PDF-Report schreiben

☒ Messung in Verlauf-CSV schreiben

ExacTrac-Nulling

ExacTrac mm / 10 zum Tischwert addieren, Vorzeichen beibehalten.

Initial: VRT -10.44 | LNG 91.74 | LAT 0.62 cm

Korrektur: VRT -0.9 | LNG +0.2 | LAT +0.3 mm -> Final: -10.53 | 91.76 | 0.65 cm

Notiz

ETD-WLT-UKL

WLT auswerten

Output öffnen

Standard-WLT Settings

Settings-Datei: \\medizin.uni-leipzig.de\data\Archiv\STR\STR-Physik\11. Scripting\Output\WLT\gui_settings.json

Analyseordner: \\s050010021\VA_TRANSFER\WLT-Images\Tb1\260616_2

Analyseordner keep: False

Output-Ordner: \\s050010021\VA_TRANSFER\WLT-Images\Tb1\260616_2

Output-Ordner keep: False

Output im Analyseordner: True

TXT: True

PDF: True

Verlauf CSV: True -> \\medizin.uni-leipzig.de\data\Archiv\STR\STR-Physik\11. Scripting\Output\WLT\wlt_history.csv

Linac: NDS-WKS-SN4632

Notiz: ETD-WLT-UKL

PDF-Report: \\s050010021\VA_TRANSFER\WLT-Images\Tb1\260616_2\wlt_report_20260625-180553.pdf (UKL-Seite ohne Signatur + Deckblatt + pylinac)

Winston-Lutz Analysis

Number of images: 4

Maximum 2D CAX->BB distance: 0.28mm

Median 2D CAX->BB distance: 0.25mm

Mean 2D CAX->BB distance: 0.22mm

Shift to Iso: facing gantry, move BB: LEFT 0.80mm; OUT 0.80mm; UP 0.10mm

Gantry 3D isocenter diameter: 0.53mm (4/4 images considered)

Maximum Gantry RMS deviation (mm): 0.28mm

Maximum EPID RMS deviation (mm): 0.32mm

Gantry-Collimator 3D isocenter diameter: 0.53mm (4/4 images considered)

Collimator 2D isocenter diameter: nicht auswertbar (nur 1/4 Bild; mind. 2 noetig)

Maximum Collimator RMS deviation (mm): nicht auswertbar (nur 1 Bild)

Couch 2D isocenter diameter: nicht auswertbar (nur 1/4 Bild; mind. 2 noetig)

Maximum Couch RMS deviation (mm): nicht auswertbar (nur 1 Bild)

Output: \\s050010021\VA_TRANSFER\WLT-Images\Tb1\260616_2

TXT: \\s050010021\VA_TRANSFER\WLT-Images\Tb1\260616_2\wlt_results_20260625-180553.txt

PDF: \\s050010021\VA_TRANSFER\WLT-Images\Tb1\260616_2\wlt_report_20260625-180553.pdf

PDF-App gestartet.

GitHub WLTopenSource

Winston Lutz Test | Auswertung

13

Maximum \ Median distance to BB (mm): 0.92 \ 0.59

Gantry 3D isocenter diameter (mm): 0.57

Shift BB to radiation isocenter (cm):

B-D: -0.077; T-G: 0.023; A-B: 0.017

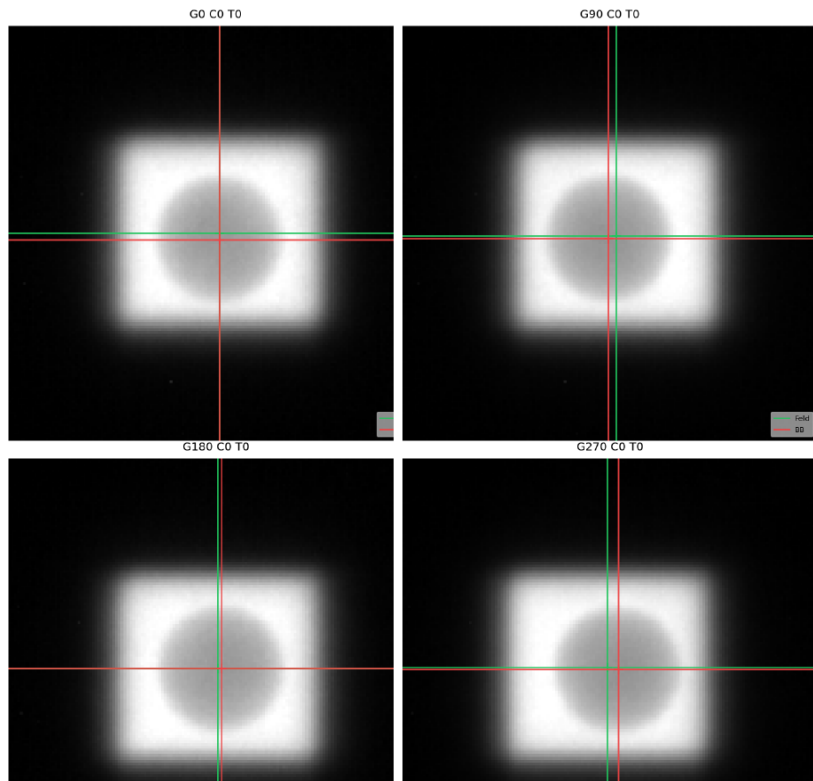
Actual couch position in IEC-61217 (cm):

VRT: -10.44; LNG: 91.74; LAT: 0.62

Recommended couch position (cm):

VRT: -10.52; LNG: 91.76; LAT: 0.64

Gantry+Collimator 3D isocenter diameter (mm): 0.57 (4/4 images considered)



Maximum \ Median distance to BB (mm): 0.28 \ 0.25

Gantry 3D isocenter diameter (mm): 0.53

Shift BB to radiation isocenter (cm):

B-D: 0.010; T-G: -0.004; A-B: -0.004

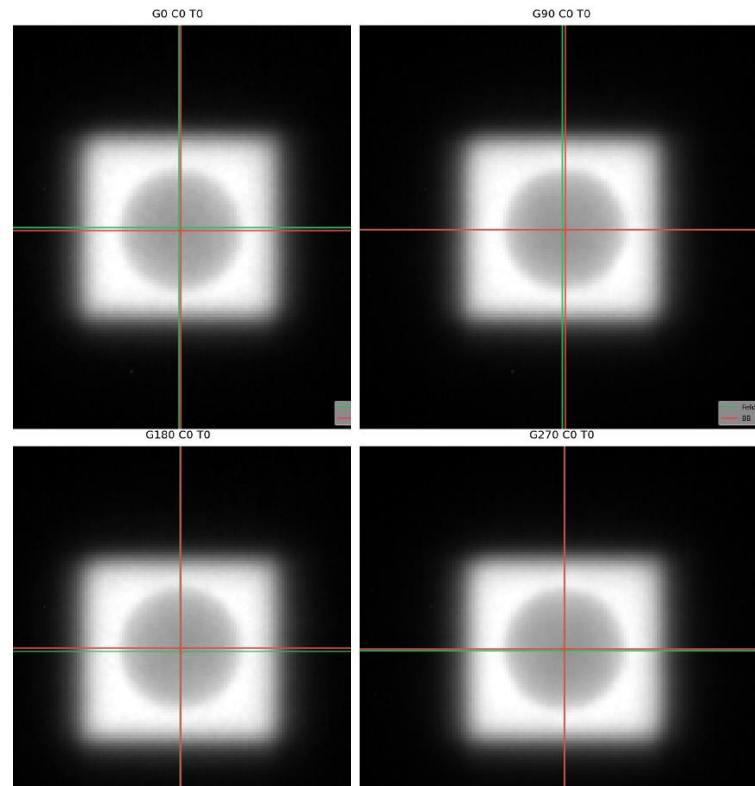
Actual couch position in IEC-61217 (cm):

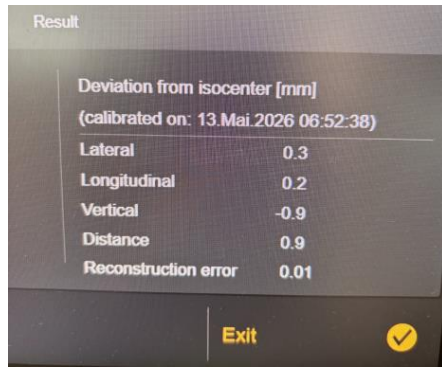
VRT: -10.52; LNG: 91.76; LAT: 0.64

Recommended couch position (cm):

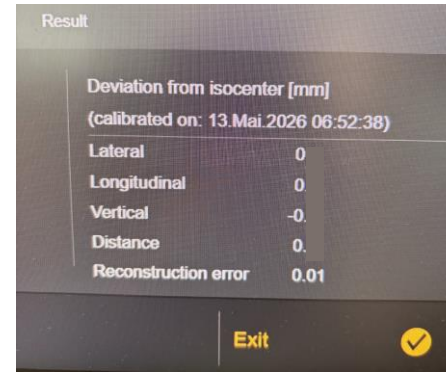
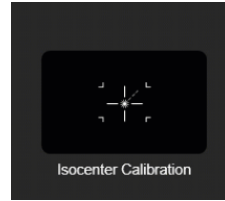
VRT: -10.51; LNG: 91.76; LAT: 0.64

Gantry+Collimator 3D isocenter diameter (mm): 0.53 (4/4 images considered)



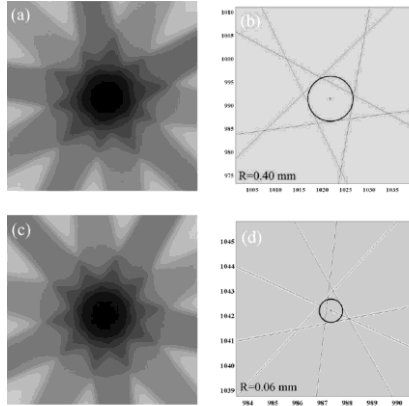


Maximum \ Median distance to BB (mm): 0.92 \ 0.59
Gantry 3D isocenter diameter (mm): 0.57

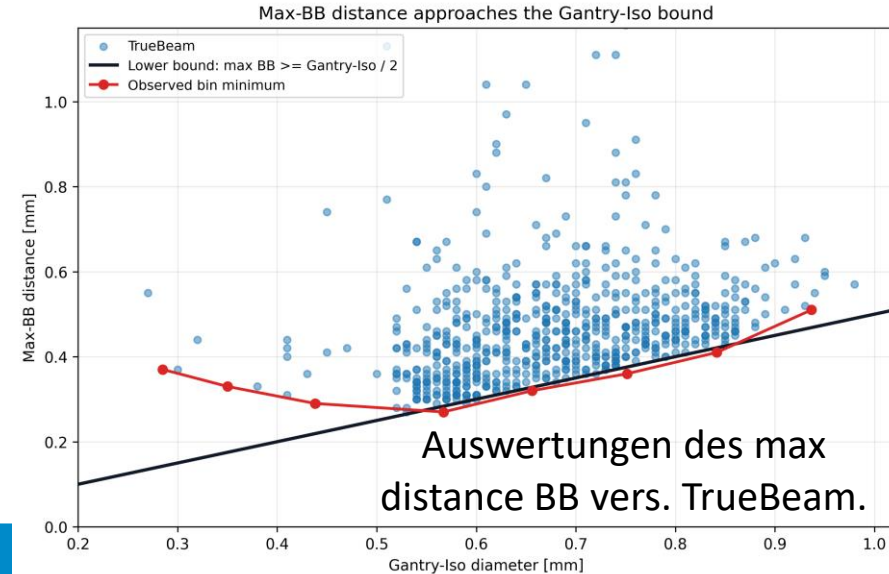


Maximum \ Median distance to BB (mm): 0.28 \ 0.25
Gantry 3D isocenter diameter (mm): 0.53

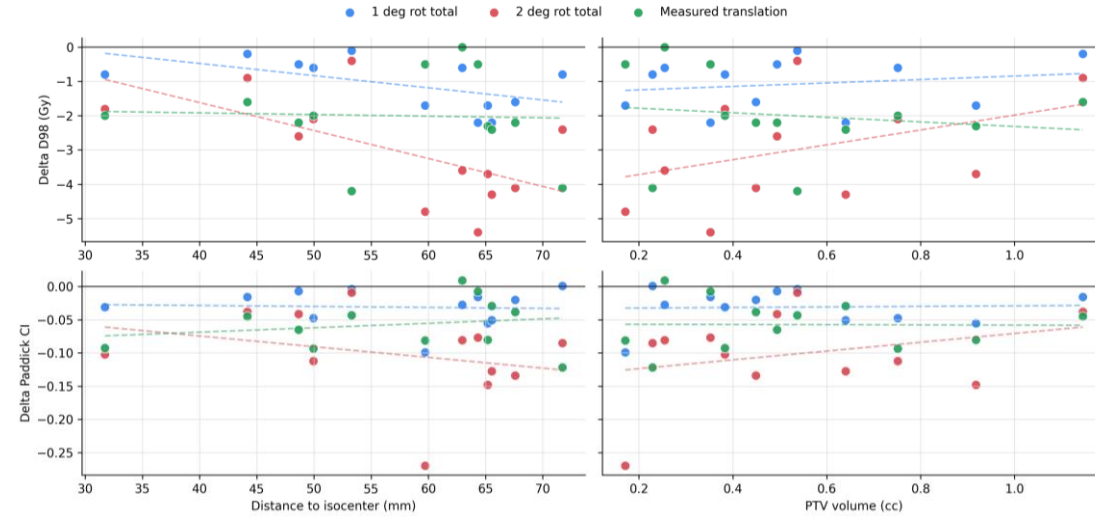
ICC-Optimierung
verringert den
systematischen
Fehler in dem
Beispiel um 0.64 mm



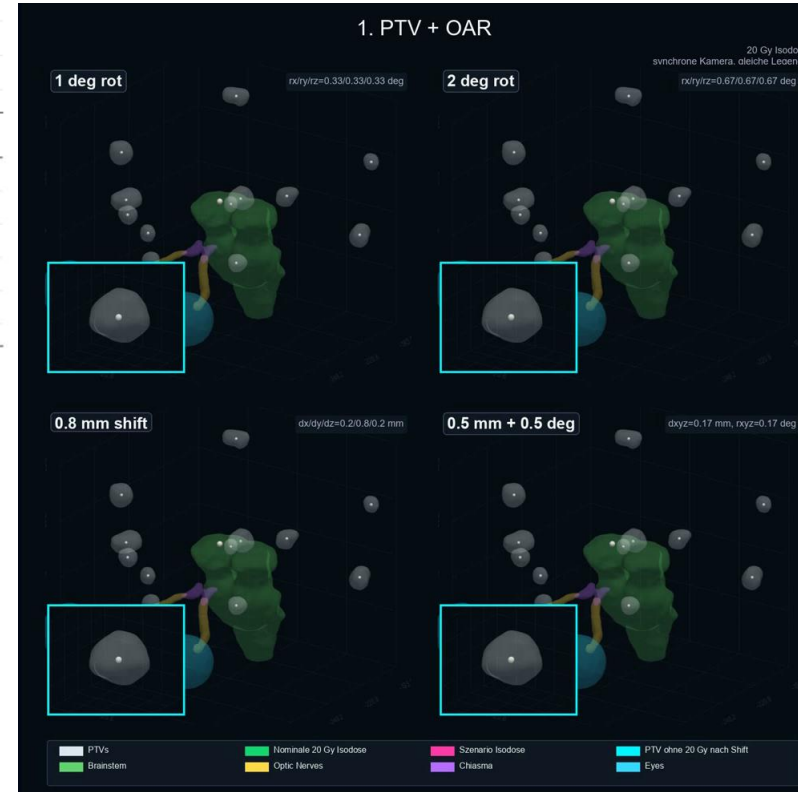
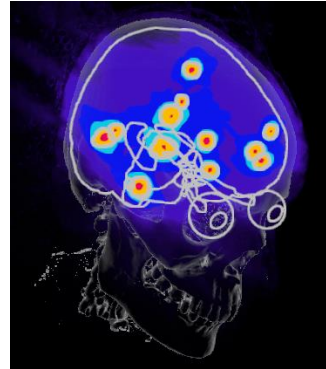
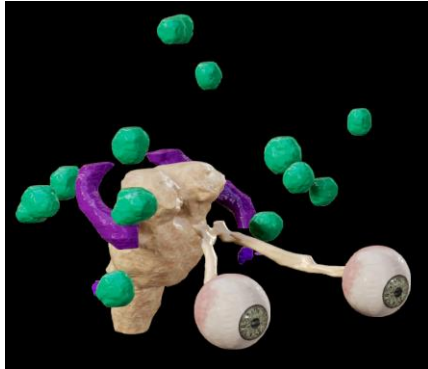
Limitierender Faktor:
Gantry-Isozentrums-
Kugel

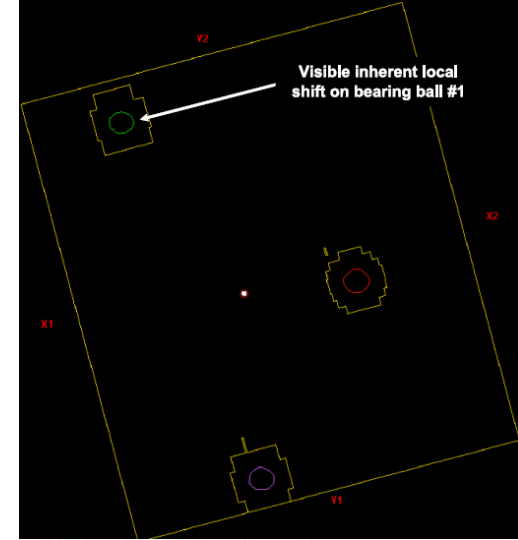
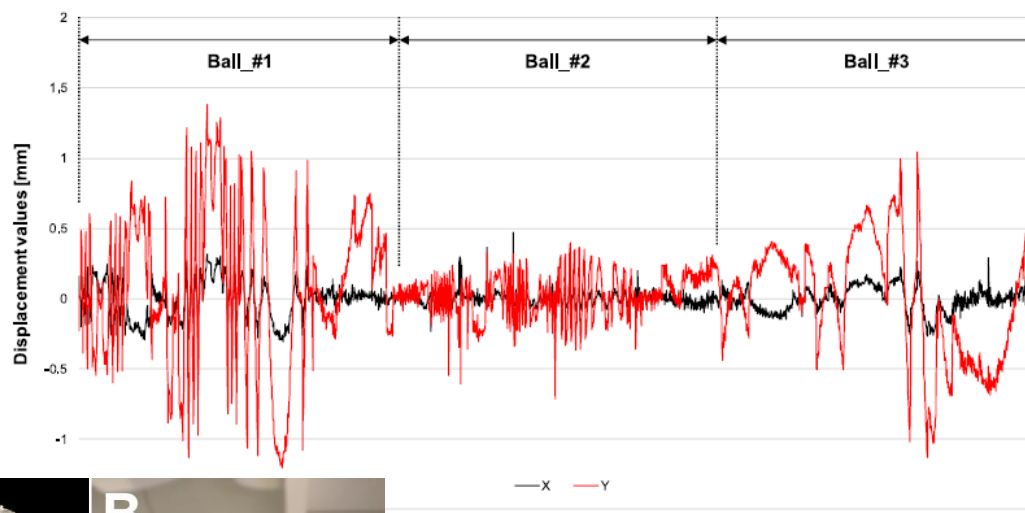


Was bedeuten „kleine“ Fehler im konkreten Fall?

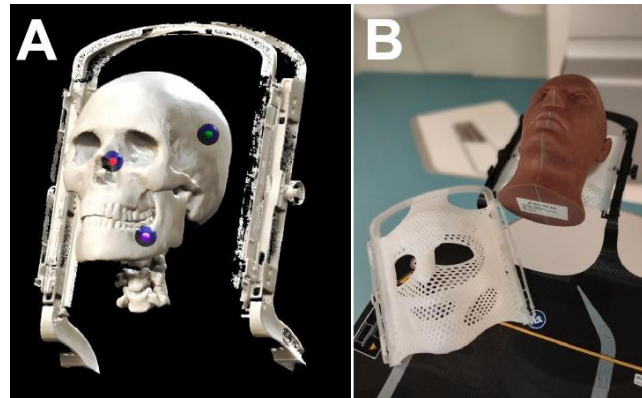


12 single targets, 20 Gy Rx; dashed lines are per-scenario linear fits.

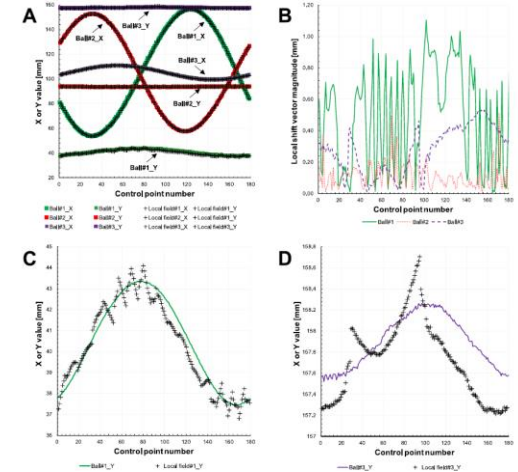




Next Step:
Off-Isocenter WLT



Retif P et al. Integrating cine EPID, dynamic delivery, and the off-axis Winston-Lutz test to enhance quality control in multiple brain metastasis stereotactic radiotherapy, Physica Medica (2024). <https://doi.org/10.1016/j.ejmp.2024.103343>



ETD-WLT-UKL | pylinac v3.44.0

WINSTON-LUTZ ANALYSE

Bereit

Standard WLT

Multi-Ball Off-Iso

Verlauf

Settings

MULTI-BALL OFF-ISO

Python-Auswertung fuer mehrere Subfelder: Feldzentrum, Kugelzentrum und Abstand je Subfeld.

RTIMAGE-Ordner

WLT-images/Output/examples/multi_ball/rtimages

keep

RTPLAN optional

LT-images/Output/examples/multi_ball/rtplan.dcm

keep

Output im RTIMAGE-Ordner speichern

Allgemeiner Output-Ordner (optional)

\\medison.uni-leipzig.de\data\Archiv\STR\STR-Phys

keep

Mets/Bild

3

Ball-Perzentil

1.0

Feldrand px

12

Messung in Verlauf-CSV schreiben

Multi-Ball auswerten

README im Explorer

Output oeffnen

ERGEBNIS UND VORSCHAU

match=Summe 82.9 deg | angle sum 82.9 deg outside tolerance

MLC-Muster:

\\s050010021\VA_TRANSFER\WLT-Images\Output\examples\multi_ball\rtimages\multi_ball_mlc_20260626-885156\67_nearest__Col_0_Ga_90_mlc.png

Feld 1: dx=-0.271 mm, dy=-0.166 mm, d=0.318 mm

Feld 2: dx=+0.204 mm, dy=-0.018 mm, d=0.205 mm

Feld 3: dx=-0.500 mm, dy=-0.253 mm, d=0.560 mm

nearest: Col 90 Ga 0 |

RI:1.2.246.352.62.1.5686798364558717472.7281764227031637411.dcm |

Bildwinkel 68.0 C52.0 T315.1 | Planwinkel 68.0 C90.0 T0.0 |

Match-Summe 82.9 deg | angle sum 82.9 deg outside tolerance

MLC-Muster:

\\s050010021\VA_TRANSFER\WLT-Images\Output\examples\multi_ball\rtimages\multi_ball_mlc_20260626-885156\68_nearest__Col_90_Ga_0_mlc.png

Feld 1: dx=-0.687 mm, dy=+0.039 mm, d=0.688 mm

Feld 2: dx=-0.042 mm, dy=-0.224 mm, d=0.227 mm

Feld 3: dx=-0.371 mm, dy=-0.036 mm, d=0.373 mm

CSV:

\\s050010021\VA_TRANSFER\WLT-Images\Output\examples\multi_ball\rtimages\multi_ball_wlt_20260626-885156.csv

JSON:

\\s050010021\VA_TRANSFER\WLT-Images\Output\examples\multi_ball\rtimages\multi_ball_wlt_20260626-885156.json

Previews: 8 Bild(er)

Graph:

\\s050010021\VA_TRANSFER\WLT-Images\Output\examples\multi_ball\rtimages\multi_ball_wlt_graph_20260626-885156.png

Bild 2/8

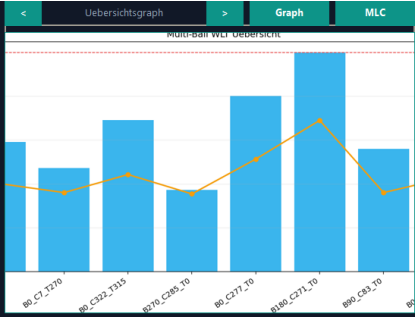
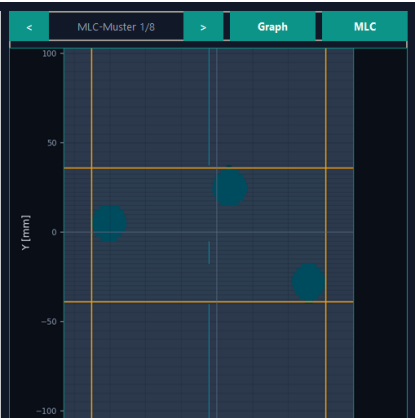
Graph

MLC

ABLAUF / README

ETD-WLT-UKL

UKL-Variante des WLT-Tools mit Standardpfaden und zwei Workflows:



Extra Modul: Multi-Ball Off-Iso



Vielen Dank für die Aufmerksamkeit.
Ich freue mich auf Ihre Fragen.

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